Chapter 3 — Circular and Rotational Motion

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. One radian is about: (a) 57.3° (b) 180° (c) 90° (d) 45°

2. Relation v and ω: (a) $v = r^2\omega$ (b) $v = r/\omega$ (c) $v = \omega/r$ (d) $v = r\omega$

3. Centripetal acceleration: (a) vr (b) v^2/r (c) v/r (d) v^2r

4. Orbital speed of satellite: (a) GM/r (b) √(gr) (c) √(GM/r) (d) √(2gr)

5. Rotational analogue of mass: (a) torque (b) angular velocity (c) angular momentum (d) moment of inertia

6. Rotational Newton's law: (a) $\tau = I\alpha$ (b) $\tau = I\omega$ (c) $\tau = L/t$ (d) $\tau = mr$

7. Angular momentum: (a) τt (b) Iα (c) Iω (d) mv

8. Near Earth orbital speed: (a) 3.0 km/s (b) 1.0 km/s (c) 7.9 km/s (d) 11.2 km/s

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is the radian a dimensionless unit?

2. Why does centripetal force do no work in uniform circular motion?

3. Why does the mass of a satellite not affect its orbital speed?

4. Why do astronauts in orbit feel weightless?

5. Why does a skater spin faster when the arms are pulled in?

6. How does moment of inertia differ from mass?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Derive the orbital speed and period of an artificial satellite. **(4)**

(b) Explain artificial gravity and derive the frequency needed in a rotating space station. **(4)**

Q.4 (a) Define moment of inertia and prove $\tau = I\alpha$ for a rigid body. **(4)**

(b) State the law of conservation of angular momentum and explain it with examples. **(4)**

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